

# Regimes' Characteristics and Time Series Forecasting

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Dei uma grande reorganizada no texto seguindo as mudanças que falamos. Na minha visão tem duas coisas separadas: uma é framework do trabalho, seja a parte teórica de definir o DGP geral e o conceito de regime-conditional metrics, seja o framework das simulações; outra é a parte aplicada, seja os DGPs, modelos, e métricas específicas que eu vou considerar, seja os parâmetros e diagnósticos das simulações realizadas. Então organizei dessa forma, mas também daria para organizar como “Theoretica framework” -> “Considered DGPs, ...” -> “Simulation framework” -> “Simulation implementation”. Coloquei as definições matemáticas no apêndice.

Várias outras mudanças, te conto na reunião. A introdução mudou mais o linguajar (learning e não inferência). O framework teorico mudou só a parte de métricas, onde adicionei definições de estacionariedade necessárias. As seções seguintes eu principalmente reorganizei como dito no comentário acima. A “Simulation implementation” é nova, e é aquela versão mais focada dos diagnósticos.

Fiz a revisão de literatura.

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# 1 Introduction

Regime switching (RS) models describe time series that exhibit different behavior – different parameters – across different regimes. They are useful to capture non-linearities in time series, having been widely used in economics and finance, for instance, to model business cycles and market volatility. There are several types of regime switching modeling, some with stochastic switching, such as Markov-switching models, and some with deterministic switching, such as threshold models.

As with any forecasting model, it is important to understand the factors that influence their performance, and how econometricians can use this knowledge to improve their models. In this work, I focus on: (i) the ability of these models to learn and generate performant forecasts under the presence of mis-specification; and (ii) how that ability relates to the characteristics of the regimes by them.

The first focus is common in forecasting econometrics: exactly identifying the data generating process (DGP) is the exception, not the rule, so the modeling goal is actually to find a robust approximator. It is important to document how each RS model behaves under different mis-specifications, explore possible universal approximators, and understand how each element of the DGP affects the learning problem of the models.

The second focus is less orthodox and specific to RS models. These models are special in the sense that they not only identify the series in question but also its states – its regimes – thus allowing the econometrician to describe the distribution of each regime and how different they are from each other. This characterization of regimes’ distributions might be informative for the model’s performance: for example, if the DGP implies different intercepts across regimes, a model whose identified regimes have the same conditional average is probably not capturing that dynamic well; or some class of model can be good at capturing that dynamic but bad at capturing changes on the persistence. These examples might seem obvious, but I will show that there is much useful information to be taken from this kind of analysis.

The nature of this project is explorative. I will simulate a diverse set of DGPs and try to find stylized facts about how each RS model adjusts to them, and how the characteristics of the estimated regimes relate to this adjustment. To make things more concrete, in the remainder of this section I synthesize the methodology, describe the patterns I hope to find, and present some of the actual findings.

## 1.1 Basic methodology and hypothesis

The methodology follows a common setup. The first step is to establish a theoretical framework that describes all RS models in a unified way. Here, I denote the separate ‘ingredients’ in an RS DGP: the *series generating process* (SGP) and the *regime generating process* (RGP). By varying these ‘ingredients’, I define a diverse set of DGPs to be considered. Then, Monte Carlo simulations are used to generate series, each being fitted by all RS models. As many questions can arise from the broad motivation of this research, creating a very general and expandable setup and implementation is a goal in itself. To restrain the focus, I will consider only stationary  $AR(1)$  processes (SGPs), and when regime changes are assumed to exist.

For the first part of the work, processing the Monte Carlo results starts with visualizing the generated series, understanding how each DGP ‘works’ and how RGP and SGP interact. With this in hand, the fit of the models can be visualized, checking which models captured the dynamics in which contexts. Then, more systematic regression analysis is done, explaining the performance of each estimated model by the DGP and model used, as well as interactions between the two, which can capture measures of mis-specification.

The second part builds on the theoretical framework. I hypothesize which characteristics of regimes can be relevant for model performance in each context, e.g., the conditional average for DGPs with intercept changes. Then, these distribution metrics are calculated for the Monte Carlo results. The processing follows the same steps as before, with initial visual diagnostics of whether the metrics indeed characterize the simulated series' regimes, and if the same pattern is found in the mis-specified models' results. Finally, regression analysis is done, now including the metrics as explanatory variables.

Aqui adicionarei o texto das perguntas mais concretas que foram pesquisadas, e os resultados.

The rest of this work is divided as follows: Section 2 presents the literature review. the general framework is presented in Section 3 and Section 4, while the specific implementation chosen is presented in Section 5. The results are split into an exploratory section (6) and a systematic one (7). Finally, Section 8 concludes.

## 2 Related literature

The regime switching literature is vast and contains many model variations. It is important to map the models, the similarities, and the differences between them. Additionally, an important starting point is to discuss what is already known about their forecasting performance and the factors that influence it. Each is done in the sections below. Before doing so, I will better define the bounds of RS literature by discussing two closely related ones.

The first is the state-space (SS) literature, with its quintessential implementation by Kalman (1960). While RS and SS models have developed as somewhat independent fields, RS can be viewed as a subset of SS, where the state (regime) variable and the observed series variable are modeled separately. This separation is central to the framework used in this paper. Bridges between the literatures include Switching State-Space Models and the seminal work by Kim (1994), which extends Hamilton’s Markov-switching model to general state-space models.

The second is the structural break (SB) literature, the most relevant starting point being Chow (1960). Much of it is devoted to diagnosing breaks, which are indeed present in RS settings, with the non-constant parameters. However, SB models typically treat breaks as exogenous and non-recurring. Bridging the gap, Bai; Perron (1998) allows for multiple unknown breaks, which can be relevant for RS contexts, while Chib (1998) demonstrates that SBs can be formulated as Markov-switching processes that have only positive probability for staying in the initial regime and switching to the next, not for switching back.

### 2.1 Regime switching models

Two of the most essential aspects of the different RS approaches are: (i) if the latent regime variable is modeled in a deterministic or stochastic fashion, and (ii) if the changes between regimes are abrupt or smooth.

The most common deterministic models are the threshold-based ones, where some observable variable being above or below some threshold(s) is what determines the regime. The work of Tong (1978) and (1980) popularized the threshold autoregressive model, each regime having its own set of autoregressive parameters. Tong proposed that capturing smooth transitions between regimes would be important, and Terasvirta; Anderson (1992), (1994) defined the smooth transition autoregressive model, where the distance between an observable variable and some threshold determines the continuous weight of each regime.

On the stochastic front, the Markov switching literature started with Hamilton (1989) via the MSAR model, where the regime is governed by an unobservable Markov process – the probability of switching to another regime is constant and depends only on the current regime. This implies a geometric distribution for the number of periods in a given regime’s instance<sup>1</sup>. The Markov switching smooth transition model, as defined by Elliott; Siu; Lau (2018), exists, but has more added complexity than the very natural jump from the TAR to the STAR model.

#### 2.1.1 Variations of the classic models

Moving forward, many variations on the regime variable modeling were created. The threshold variable can have a delay or some transformation; it can be the series itself, an exogenous variable, or even a non-linear combination of variables (Chen; Liu; So, 2011). The probability distributions of MS models

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<sup>1</sup>Throughout this document, ‘regime instance’ will be used to describe a contiguous period of time without switches. In a given series, a given regime can have several instances.

were extended to allow different distributions for the time spent in one regime, and dependence on more past values (Ferguson, 1980). The smooth transition function has several options, with common ones being the logistic and the exponential. The models were generalized to any number of regimes, with the STAR model being equivalent to a Neural Network, as described by Medeiros (2000).

Blurring the line between deterministic and stochastic models, Chang; Choi; Park (2017) uses threshold dynamics but adds an innovation that is dependent on the previous state’s innovation, and simplifies to a MS model when the threshold dynamic is exogenous and stationary. Wu; Chen (2007) creates a half-threshold, half-random regime process. There are also unsupervised approaches to estimation, that make no assumption on the nature of the latent process, as by Akioyamen; Tang; Hussien (2020), where some clustering model can be used to identify regimes, and later the functional form can be estimated for each regime separately.

As I will note in this work, the functional form across regimes and the regime process itself are fairly independent, thus other variations arise from considering more complex functions than the autoregressive one. ARMA models have their RS counterparts (Brockwell; Liu; Tweedie, 1992). Not only the mean, but the variance can also be modeled: the ARCH/GARCH family, very relevant for finance, have their regime switching versions (Chen; Liu; So (2011)). More recently, models such as decision trees have been adapted to the regime switching context, as by Adam; Ötting; Michels (2024). Similarly, there are also models for vectors of time series.

General reviews on RS models include (Tan; Wu, 2025), (Potter, 2000), and (Hamilton, 2020), while (Chen; Liu; So, 2011) focuses on threshold models, (Dijk; Teräsvirta; Franses, 2002) in smooth transition, and Song; Woźniak (2021) in Markov switching. Note that RS models are considered in both frequentist and Bayesian frameworks, with the latter inheriting a lot from the SS models estimation literature.

## 2.2 Forecasting performance

There are many research topics in RS performance. I’ll focus on (i) important factors that relate to model selection and hyperparametrization, to contextualize the decisions I made in this work; and (ii) comparisons between models, to contextualize the experiments I ran.

### 2.2.1 Hyperparametrization

While RS models are frequently cited for their superior in-sample fit, which is useful for explaining historical phenomena, Dacco; Satchell (1999) noted that even minor errors in forecasting the future regime state can propagate through the non-linear structure, causing the overall prediction to perform worse than linear alternatives. Furthermore, standard metrics like mean squared error may be ill-suited for evaluating non-linear time series, potentially masking the utility of these models in capturing turning points or specific economic states [4].

A primary challenge in RS modeling is managing the trade-off between flexibility and overfitting. The most critical decision is on the number of regimes: too few can underfit, while too many might lead to overparameterization. Similarly, allowing all parameters to switch can help capture complex dynamics and avoid mis-specification, but doing it when unneeded often dilutes out-of-sample power (Tan; Wu, 2025).

Each model also has its own specificities. For Markov Switching models, the estimation method is relevant: for example, EM algorithms have been noted to balance accuracy and speed in high-dimensional settings (Akbal, 2024). Moreover, the translation of soft posterior probabilities into hard regime labels affects accuracy, and different rules have different properties (Hall; Pagan, 2025). For deterministic models, the challenge lies in variable selection: identifying the correct threshold

variable, delay parameter, or non-linear combination of variables remains a significant hurdle for effective specification.

### 2.2.2 Comparisons between models

Many papers compare the different RS models in many different contexts (Clements; Krolzig, 1998), (Bierbrauer; Trück; Weron, 2004), (Pinson et al., 2008), (Janczura; Weron, 2010), (Elias; Wahab; Fang, 2014), (Chen; Bunn, 2014), (Panopoulou; Pantelidis, 2015), (Verne, 2021), (Aydın; Mermi, 2022). No single model is universally superior, the same context can present a different “best” model depending on the focus (e.g. nowcasting, regime identification, portfolio performance, etc.) (Akbal, 2024).

TAR models are best employed when regime changes are triggered by a single, observable variable with rigid boundaries. They’ve shown effective for financial assets like gold prices and exchange rates, where transitions are fast rather than gradual (Aydın; Mermi, 2022). However, their reliance on observable triggers is a limitation: in contexts like offshore wind power, where fluctuations are driven by complex, non-observable states, TAR models fail to capture the underlying dynamics and significantly underperform compared to latent variable models (Pinson et al., 2008).

STAR models are theoretically appropriate for gradual economic adjustments but often face practical identification challenges. In many financial applications, the estimated smoothness parameter becomes so high that the model collapses into an abrupt threshold model, rendering the specific “smooth” specification inefficient (Aydın; Mermi, 2022). However, STAR models can outperform MS in macroeconomic contexts characterized by explosive volatility, such as GDP growth requiring the capture of “brutal” transitions typical of recession phases (Verne, 2021).

Markov-Switching Models (MS/MSAR) are the superior choice when regimes are driven by latent, multi-factor variables (e.g., market sentiment or meteorology) rather than a single observable index. But, with a more flexible regime framework, they have shown to be more sensitive to specification of the number of regimes Janczura; Weron (2010).

### 3 Theoretical framework

In this section, I define the theoretical framework that guides the rest of this work. First, I define the general structure of RS DGPs, aligning all in a common mathematical representation, and relate the concepts of models and metrics to it. An important idea is the separation of the DGP into RGP and SGP.

#### 3.1 The general regime switching DGP

Let  $y_t \in \mathbb{R}$  denote the series of interest at time  $t \in 1 : T^2$ ,  $T \in \mathbb{N}$ . Let  $S \in \mathbb{N}$  denote the number of regimes. The *regime variable* is a vector  $r_t \in \mathbb{R}^S$  of ‘weights’ for each regime, indexed by  $r_t^s$ ,  $s \in 1 : S$ .

In this work, I consider only univariate series.

A DGP can be written in terms of a pair: *regime generating process* (RGP) and *series generating process* (SGP). This is essentially the separation between the state/system equation, and the output/measurement equation, in state space models. They are functions with parameters  $\Theta_r$  and  $\Theta_y$ , respectively, such that:

$$\begin{aligned} r_t &= \text{rgp}(y_{1:(t-1)}, r_{t-1}, t; \Theta_r) \\ y_t &= \text{sgp}(y_{1:(t-1)}, r_t, t; \Theta_y) \\ &= \text{sgp}(y_{1:(t-1)}, \text{rgp}(y_{1:(t-1)}, r_{t-1}, t; \Theta_r), t; \Theta_y) \end{aligned} \quad (3.1)$$

Without loss of generality, I restrict the regime weights to be non-negative and sum to one, i.e.  $r_t \in [0, 1]^S$  and  $\sum_{s=1}^S r_t^s = 1$ . Notably, the number of regimes  $S$  is a parameter in  $\Theta_r$ , and  $\Theta_y$  is actually a set of different parameters for each regime, each indexed by  $\Theta_y^s$ . This means that the SGP can be written as:

$$\text{sgp}(y_{1:(t-1)}, r_t, t; \Theta_y) = \sum_{s=1}^S f_{\text{sgp}}(y_{1:(t-1)}, t; \Theta_y^s) \cdot r_t^s \quad (3.2)$$

Note how each regime is weighted by the regime variable  $r_t^s$ . In the simplest case, this weight is binary ( $r_t \in \{0, 1\}^S$ ) – only one of the regimes is ‘turned on’, and all the others are ‘turned off’. However, in some models, such as Smooth Transition, the weights can be continuous, with different regimes being partially ‘on’ at the same time.

Furthermore,  $\Theta_y$  could encode different functional forms for each regime. As this is not common, I refer to  $f_{\text{sgp}}$  as the *SGP functional form*, or simply *SGP*, and the set of parameters  $\Theta_y$  as the *regime nature*, as they define what actually changes between each regime. Figure 1 illustrates this structure.

Figure 1: The general RS DGP structure.

$$\begin{array}{c} \text{DGP} = \underbrace{\sum_{s=1}^S \text{sgp}(\cdot; \Theta_y^s) \cdot r_t^s}_{\text{SGP}} \quad \& \quad \underbrace{r_t = \text{rgp}(\cdot; \Theta_r)}_{\text{RGP}} \\ \underbrace{\hspace{10em}}_{\text{functional form} \quad \& \quad \text{regime nature}} \\ \underbrace{\hspace{10em}}_{f_{\text{sgp}} \quad \& \quad \{\Theta_y^1, \dots, \Theta_y^S\}} \end{array}$$

<sup>2</sup>Let  $a : b := \{a, a + 1, \dots, b\}$  for  $a \leq b \in \mathbb{Z}$ , and  $y_{a:b} := \{y_a, \dots, y_b\}$ .



To construct a diverse set of DGPs, I combine different RGP, SGP, and regime natures. One of the challenges of this work is to choose a comprehensible set of these elements, and analyze their differences in a systematic but manageable way.

In the notation above I omitted the error term inside  $f_{\text{sgp}}$ . Many distributions are interesting, specially fat-tailed and skewed ones. The parameters  $\Theta_y$  can even encode different error distributions across regimes. However, if we constrain the same distribution across regimes, with the possible exception of a multiplicative factor, we can simplify the notation and implementation, writing the DGP as a function that receives a sequence of random errors and returns the series and the regimes:

$$(y_{1:T}, r_{1:T}) = \text{dgp}(\varepsilon_{1:T}; \Theta_r, \Theta_y) \quad (3.3)$$

Consider the notation shorthand  $y := y_{1:T}$ , and similarly for other variables, used for the rest of this work.

Let the set of considered DGPs be  $P$  (for ‘processes’). These are present in the literature, as discussed in Section 2, and will be defined in Section 5.

### 3.2 Models

Consider a model  $\text{mod}$  as a function with (hyper-) parameters  $\Theta_m$  that generates the fitted values and  $H$ -step ahead predictions of the series and regimes. The model can also return a set  $\hat{\pi}$  of general metadata, e.g. the estimated coefficients.

$$(\hat{y}, \hat{r}, \hat{\pi}) = \text{mod}(y_{1:(T-H)}; \Theta_m) \quad (3.4)$$

Notably, the number of estimated regimes  $\hat{S}$  is a parameter in  $\Theta_m$ , which may or may not be equal to  $S$ .

Let the set of models be  $M$  (for ‘models’). Also present in the literature, they will be defined in Section 5.1.

### 3.3 Regime conditional metrics

A regime-conditional (RC) metric  $\text{met}$  is function that receives a vector of series and a vector of regimes, and returns a sequence with one value for each regime. They are used to characterize the distribution of  $y_t$  or  $(y_t, y_{t-j})$  within each regime.

$$\text{met} : (y, r) \mapsto \mathbb{R}^S \quad (3.5)$$

An example is the function that returns, for each regime  $s$ , the mean of the series weighted by  $r_t^s$ . This can be done for many common metrics, and is equivalent to mapping  $(y, r)$  to the  $S$  sets  $R_s$  of regimes’ observations<sup>3</sup>, then applying the metric to each set. The benefit of the first approach is that it is more general, allowing for non-binary – i.e., smooth transition – regimes.

For the joint distribution  $(y_t, y_{t-j})$ , the metrics are more complex, as they must consider only the windows  $(y_t, \dots, y_{t-j})$  fully contained in the same regime instance. This is further described in Section A.2.

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<sup>3</sup> $R_s := \{y_t : r_t^s = \max\{r_t\}\}.$

In any case, RC metrics may lump together observations from different time windows. For them to describe a well-defined regime distribution, it is required that the series be stationary within each regime. This will impose restrictions on the DGPs that this work will consider.

### 3.3.1 Within-regime stationarity

For the distribution of a regime to be well defined, all datapoints within it must be drawn from the same distribution. Formally, *within-regime strong stationarity* requires, for all regimes  $s \in S$ , a CDF  $F_s$  such that:

$$F_s(y_{t_1}, \dots, y_{t_n}) = F_s(y_{t_1+j}, \dots, y_{t_n+j}) \quad \forall \{y_{t_1}, \dots, y_{t_n}\}, \{y_{t_1+j}, \dots, y_{t_n+j}\} \subset R_s, \quad \forall j \in \mathbb{N}_* \quad (3.6)$$

As we intend to characterize the distributions with specific metrics, weaker assumptions can be made. If we restrict ourselves to the moments of the (joint) distribution, we can require the weak version. Formally, *within-regime weak stationarity* requires<sup>4</sup>, for all  $s \in S$ , that:

$$\begin{aligned} E[y_t] &= E[y_{t'}] & \forall y_t, y_{t'} \in R_s \\ \text{Var}[y_t] &= \text{Var}[y_{t'}] & \forall y_t, y_{t'} \in R_s \\ \text{Cov}[y_t, y_{t-j}] &= \text{Cov}[y_{t'}, y_{t'-j}] & \forall \{y_t, \dots, y_{t-j}\}, \{y_{t'}, \dots, y_{t'-j}\} \subset R_s, \\ & & \forall j \in \mathbb{N}_* \end{aligned} \quad (3.7)$$

Processes that have a non-binary  $r_t$ , i.e., smooth transitions, do not have truly separated regimes, and thus, generally do not satisfy the conditions above. Thus, the metrics cannot be interpreted as, e.g., “the mean of all datapoints in a regime”. Still, their information might be useful, as will be studied in this work.

### 3.3.2 Aspects of RC metrics usage

There are two important aspects of the RC metrics usage. First is whether to use the whole sequence of values for each  $s$ , or to condense it into a single value of dispersion across regimes. An example of the latter is the ‘average pairwise distance between the RC means’, a single value that describes how distant the levels of the regimes are. This is equivalent to composing a dispersion function  $\text{disp}$ :

$$\text{disp} \circ \text{met} : (y, r) \mapsto \mathbb{R}^S \mapsto \mathbb{R} \quad (3.8)$$

Second, which series to use: the true or estimated ones. One can use the true values  $(y, r)$  and get the characteristics of the true DGP, or the estimated values  $(\hat{y}, \hat{r})$  and get the characteristics of the estimated model<sup>5</sup>. Another option is to calculate the difference between the former and the latter<sup>6</sup>. Another option is to calculate the metric of the difference  $(y - \hat{y}, r)$  or  $(y - \hat{y}, \hat{r})$ .

This framework allows for mixing and matching these options, each being useful to answer different questions. In this work, I focus on the estimated series, as they are the only thing available to the econometrician in practice, and on using the dispersion of RC metrics, as it is more comparable across DGPs and models.

<sup>4</sup>The ACF condition should be read as “ $j$ ’th autocorrelations between time-points within the same regime instance should always be equal”, not as “ $j$ ’th autocorrelations of every time-point in the same regime instance should always be equal”, although the latter is true for AR processes with order higher than  $j$ .

<sup>5</sup>Note that the value of  $S$  and  $\hat{S}$  can be different, and thus, so the dimension of the metric’s output.

<sup>6</sup>This is only possible if  $S = \hat{S}$  and there is an unambiguous way to match the estimated and true regimes.

Não sei se foi uma tangente muito grande falar dessas outras opções que não vou usar, ainda mais porque pode não estar claro pro leitor qual tipo de pergunta cada ajuda a responder. Eu gostaria de informar essa flexibilidade do framework, mas talvez valha mais colocar isso num apêndice.

Less generally, sometimes there are other possible estimators for the same population RC metric, instead of simply using  $(\hat{y}, \hat{r})$ . A special case is when the metric is a moment of the (joint) distribution, and the SGP is simple: one can simply plug the estimated parameters into the analytical formula for the moment, and generally have a better estimator. This is further discussed in Section [A.2](#).

Let the set of metrics  $(\text{disp} \circ \text{met})$  be  $C$  (for ‘criteria’). These will be defined in Section [5.4](#), but are mostly based on the moments of  $y_t$  and of the pair  $(y_t, y_{t-j})$ ,  $j \in \mathbb{N}$ , and the performance metrics for the dependent variable.

One can also be interested in describing the RGP, with information such as the average duration of each regime instance, the transition probabilities and measures derived from it, amongst others. I’ll use these as control variables in the regression analysis.

## 4 Simulation framework

One of the partial goals of this work was to create the theoretical framework described in the last section in a very general and expandable way, that easily allows for different exercises, even if they are not considered here. Similarly, the simulation structure was designed to follow the same concept.

There are the following steps to perform the simulations:

1. Generate random errors for all the DGPs.
2. For each DGP and simulation, fit and predict the model, generating  $(y, r)$ .
3. For each DGP, simulation, and model, obtain  $(\hat{y}, \hat{r})$ .
4. For each DGP, simulation, and model, compute each metric.
5. Aggregate the metrics, performance information, and DGP and model descriptors into a dataset.

### 4.1 Forecast horizon

For the forecast performance, I focus on 1-step ahead predictions. It would be interesting to expand that, be it with locally-projected models or not.

To obtain more than one prediction per simulation, I simulate a  $T - H$ -long series, and obtain  $H$  predictions. There are two possible approaches:

1. For each iteration  $h \in 1 : H$ , the model is estimated with the window  $h : (T - H + h - 1)$ , and generates  $\hat{y}_{T-H+h}$ .
2. The model is estimated once with the window  $1 : (T - H)$ , then for each  $h$ ,  $\hat{y}_{T-H+h}$  is generated using  $y_{1:(T-H+h-1)}$ .

The second approach is computationally cheaper, allowing for more simulations and DGPs to be considered. It is the one used in this work, but note that it is less accurate to what would be done in practice, as econometricians often re-estimate their models with new data.

### 4.2 Simulation hyperparameters

The hyperparameters of the simulation are as follows:

- Number of simulations:  $I$ . Its main effect is on the the precision of the results, and diversity of series.
- Forecast horizon:  $H$  predictions of 1-step ahead values. Also affects the precision of the results, but does not change the diversity of series.
- Total number of observations:  $T$ . Its main effect is on the ability of the models to learn the dynamics and separate the regimes. Results for higher  $T$ 's are more relevant for contexts with a lot of data, such as high-frequency financial data, while lower  $T$ 's are more relevant for contexts with less data, such as macroeconomic data.
- Burn-in period:  $B$ . Its main effect is on reducing the dependence of the initial values, but with stationary processes, this is not too problematic.

Let  $i \in 1 : I$ ,  $I \in \mathbb{N}$  be the simulation index.

### 4.3 Simulating series

I will only consider DGPs have the same error distribution – but note that a DGP can have a volatility parameter multiplying its error. For each DGP, indexed by  $p \in 1 : |P|$ , there are  $I$  random error vectors created, each of size  $T$ . Let  $E$  denote the set of all errors. Let  $E_{p,i}$  denote the vector of errors generated for the  $p$ -th DGP and the  $i$ -th simulation. Similar indexing definitions will be used for similar collections throughout this document.

Let  $Y$  and  $R$  denote the sets of generated series and regime variables. They are computed given  $E_{p,s}$ :

---

```

1: Initialize  $Y$  and  $R$ 
2: for  $p$  in  $1 : |P|$  do
3:   for  $i = 1$  to  $I$  do
4:      $Y_{p,i}, R_{p,i} \leftarrow P_p(E_{p,i})$ 
5:   end for
6: end for

```

---

### 4.4 Estimating models

Now, for each simulation, I estimate each model, generating the sets  $\hat{Y}$ ,  $\hat{R}$ , and  $\hat{\Pi}$ . The models are trained using only  $y_{(B+1):(T-H)}$ , to avoid the burn-in period and leave space for the forecast horizon.

The nesting order is the same as above, for consistency, but with an additional inner loop for the model estimation.

---

```

1: Initialize  $\hat{Y}$  and  $\hat{R}$ 
2: for  $p$  in  $1 : |P|$  do
3:   for  $i$  in  $1 : I$  do
4:     for  $m$  in  $1 : |M|$  do
5:        $\hat{Y}_{p,i,m}, \hat{R}_{p,i,m}, \hat{\Pi}_{p,i,m} \leftarrow M_m(Y_{p,i}, R_{p,i})$ 
6:     end for
7:   end for
8: end for

```

---

### 4.5 Calculating metrics

Then, for each model, the dispersion of the RC metrics are calculated and stored as columns of a dataset  $D$ . Each row of  $D$  is identified by  $(p, i, m)$ .

---

```

1: Initialize  $D$ 
2: for  $(p, i, m)$  in  $(1 : |P|) \times (1 : I) \times (1 : |M|)$  do
3:   for  $c$  in  $1 : |C|$  do
4:      $D_{(p,i,m), c} \leftarrow C_c(\hat{Y}_{p,i,m}, \hat{R}_{p,i,m})$ 
5:   end for
6:    $\hat{\Pi}_{p,i,m}$  is appended to  $D_{p,i,m}$ .
7:   Categorical variables  $(p, i, m)$  are appended to  $D_{p,i,m}$ .
8: end for

```

---

Recall the discussion in Section 3.3 about the two different aspects of RC metrics usage. With different choices regarding the usage of true or estimated series, the function  $C_c$  could receive different inputs (e.g.  $Y$  and  $R$ ). Additionally, the function could return the whole sequence of RC metrics, not a single value, then, each row would be identified by  $(p, i, m, s)$ .

The dataset  $D$  is already in a friendly format for analyzing the relationship between the performance of each observation and the characteristics of the regimes, as well of considering stratifications by DGP and model.

## 5 Implementation

The framework described in the last two sections is very general, and allows for a lot of different exercises. In this specific work, I focus on a specific set of DGPs, models, and metrics. These are described here.

First of all, I focus only on DGPs where there are regime switching (or structural breaks), and specifically two regimes ( $S = 2$ ). More information about the ability of the models to identify regime dynamics with different (or zero) number of regimes is an interesting topic.

The choices of hyperparametrization were made to balance the ‘population’ of DGPs. Each of the four RGP have equal representation, each with a symmetric and an asymmetric variation. There are two RNs for each of the three  $AR(1)$  parameters, a big and a small change. The related hyperparametrizations were chosen guided by the concept of “regime separation”, described in Section 6.1.

The choice of models and their hyperparametrization is more flexible, as they do not affect the ‘population’ of the experiments. Each of the RGPs’ empirical model counterparts is used, with a ‘generic’ hyperparametrization. But it would be interesting to increase the diversity of models.

The metrics are limited to the most essential descriptors of the regime distributions, the 1st, 2nd moments, and the lag 1 autocorrelation. This is another set that could be expanded easily. Performance and RGP-related metrics were also defined for the regression analysis.

Finally, some diagnostics on the series generation and model estimation are included.

### 5.1 Considered SGPs

The functional form of the SGP could be important in its interaction with the other ingredients of the DGP. Additionally, some topics are interested in specific SGPs, such as conditional volatility in finance and GARCH models. For now, however, this does not seem to be the main point of interest. I will consider only an  $AR(1)$  process, for its simplicity, popularity, and ease of estimation.

Additionally, I’ll only consider a Gaussian distribution for the error term, ignoring fat-tailed and skewed distributions. The distribution is regime-invariant, except for the multiplicative variance parameter  $\sigma$ .

As discussed, it is useful to consider only within-regime weak stationarity, even though many interesting DGPs are non-stationary. This restricts the absolute value of the  $AR(1)$  parameter to 1. The only SGP functional form considered is the following:

$$\begin{aligned} f_{\text{sgp}}(\cdot; (\mu^s, \rho_1^s, \sigma^s)) &= \mu^s + \rho_1^s y_{t-1} + \sigma^s \cdot \varepsilon_t \\ \varepsilon_t &\sim \mathcal{N}(0, 1) \\ |\rho_1^s| &< 1, \quad \sigma^s > 0, \quad \forall s \in 1 : S \end{aligned} \tag{5.1}$$

Several others SGP’s could be considered, such as ones with transformations of  $y_t$  as regressors, non-linear regression forms, or even decision trees, as in the common model Markov-switching Random Forest. Still, the  $AR(1)$  is an essential building block, and its simplicity helps isolate the effects of the other ingredients.

## 5.2 Considered RGPs and models

The next ‘ingredient’ is the RGP. I will consider the options Self-Exciting Threshold (SET), Smooth-Transition (ST), and Markov-Switching (MS). Structural Break (SB) is included to study how RS models perform in the case of breaks without reoccurring regimes.

Each of these RGPs has empirical model counterparts, which are also considered. There is an additional model with an unsupervised approach where the regimes are defined by some clustering technique and each regimes’ AR is estimated independently afterwards (Clustering + AR, CAR). Finally, a non-RS Random Forest (RF) model is included as a benchmark.

The formal definition of each RGP/model is presented in Section A, first the RGP hypothesis, then the empirical model’s estimation strategy.

For all RGPs, an option with equally likely regimes and an asymmetric variation is considered.

- **Structural Breaks:**

- A single break at  $T/2$ , and a single break at  $2T/3$ .

- **Self Exciting Threshold:**

- Fixed hyperparameters: switching based on  $y_{t-1}$ . Different lags are often specific to timing-related issues, and not considered here.
- A single threshold at 0, and a single threshold at 0.5.

- **Smooth Transition:**

- Fixed hyperparameters: switching based on  $y_{t-1}$ , logistic’s CDF as transition function.
- A single threshold at 0, and a single threshold at 0.5.

- **Markov Switching:**

- Symmetric matrix, high persistence ( $P(s|s) = 0.9$ ), symmetric matrix, low persistence ( $P(s|s) = 0.6$ ).
- Asymmetric matrix, high persistence ( $P(1|1) = 0.9$ ,  $P(1|2) = 0.7$ ), asymmetric matrix, low persistence ( $P(1|1) = 0.8$ ,  $P(1|2) = 0.6$ ).

Tem várias outras parametrizações já feitas no código e com o texto escrito, mas deixei só essas aqui caso algo mude. Alguma sugestão de formatação melhor do que a atual?

For the models, most hyperparameters are as follows:

- All the coefficients are assumed to change across regimes, as this is a common assumption, especially in the face of possible mis-specification.
- The number of regimes  $\hat{S}$  is fixed, not estimated. Models are estimated with 2 regimes.
- The values of model-specific hyperparameters are the same as the related RGP’s values.

## 5.3 Considered regime natures

The following regime natures are considered, each representing a different way in which the SGP parameters change across regimes:

- **Mean ( $\mu$ ) change:**

- Small difference: ( $\mu^1 = 0$ ,  $\mu^2 = 0.5$ )
- Large difference: ( $\mu^1 = 0$ ,  $\mu^2 = 2$ )



- **Persistence ( $\rho_1$ ) change:**
  - Small difference: ( $\rho_1^1 = 0.6, \rho_1^2 = 0.4$ )
  - Large difference: ( $\rho_1^1 = 0.9, \rho_1^2 = 0.1$ )
- **Volatility ( $\sigma$ ) change:**
  - Small difference: ( $\sigma^1 = 1, \sigma^2 = 2$ )
  - Large difference: ( $\sigma^1 = 1, \sigma^2 = 4$ )

Idem, existem outras opções. Alguma sugestão de formatação melhor do que a atual?

Note that the regimes are always ordered increasingly by the parameter of interest. In general, the large vs. small differences will be interesting to analyze in relation to each other. To compare different types of changes, only the large differences will be considered, for simplicity.

## 5.4 Considered metrics

The goal with RC metrics is to capture the change in the series characteristics across regimes. One important option is the estimated parameters of the model for each regime, e.g.,  $(\hat{\rho}_s)_{s \in \hat{S}}$ ,  $(\hat{\mu}_s)_{s \in \hat{S}}$ , etc. One might think that this would outshine all other metrics, but in more complex cases where more than one parameter changes, this becomes less useful. More general metrics generate benefits from their abstraction over the DGP. Additionally, in simple SGPs, there often is a metric that is directly connected to changes in parameters, such as the conditional average for changes in intercept.

In this work, I focus on the moments of the distribution of  $y_t$  and  $(y_t, y_{t-j})$ . Specifically, the RC metrics considered are the RC mean, RC standard deviation, and RC autocorrelation of lag 1. Higher lags could be considered, but in the simple  $AR(1)$  context this would bring little additional information.

Talvez o 3ro e 4to momentos sejam interessantes, especialmente o 3ro porque alguns DGPs geram séries assimétricas.

As stated before, the RC mean and RC SD are simply the mean and SD of each set  $R_s$ . The autocorrelation is similar, but must be calculated separately for each concurrent set of observations in  $R_s$ . The formal definitions are stated in the Section A.2.

As the focus is on the dispersion of RC metrics, two important measures to consider are the standard deviation and the average pairwise absolute difference. For only two regimes, they are very similar and the absolute difference is more intuitive. All the metrics are composed such that all  $d \circ c \in C$  return a single real value, and  $d(x) = |x_1 - x_2|$ .

There are some possible expansions on this work's metrics calculation. One is to use non-standard weights for the empirical moments, giving more importance to observations near the edges of regimes' instances. Another is to use a cluster separation measure, such as the silhouette score, instead of a simple absolute distance between the RC metrics. Finally, one can use distribution distance metrics, such as the Earth Mover's Distance, on the empirical distribution of each regime.

The list of considered metrics is as below:

- First moment: RC mean  $\hat{\mu}(y|S)$ .
- Second moment: RC standard deviation  $\hat{\sigma}(y|S)$ .
- First autocorrelation: RC 1st autocorrelation  $\hat{\rho}_1(y|S)$ .

### 5.4.1 Performance and RGP metrics

The performance metrics considered are  $R^2$  for fit performance, and RMSE and MAPE for forecasting performance. The MSE is not included, following Dacco; Satchell (1999).

Other metrics pertaining to the RGP will be included as controls in the regression analysis: the number of regime switches divided by  $T$ , as a measure of switching frequency; the absolute difference between the average duration of regime 1's instances and regime 2's instances, as a measure of regime asymmetry.

For works with more than two regimes, more complex measures of regime asymmetry can be used, ones that consider the whole matrix of transition probabilities.

## 5.5 Simulation implementation

The implementation of the simulations, as well as their analysis in the next sections, is done with the R programming language, and the code can be found in [this paper's repository](#). The code is highly modular and fully reproducible, following the intent of setting up an expandable framework.

Following Section 4.2, the chosen hyperparameters are as below. Some values are lower than they could be due to computational constraints.

- Number of simulations:  $I = 500$ .
- Forecast horizon:  $H = 10$  predictions of 1-step ahead values.
- Total number of observations:  $T = 100$ .
- Burn-in period:  $B = 5$ .

Os valores finais podem mudar.

The error sequences were generated in parallel, using `rTRNG::rnorm_trng`. The models were estimated with `mbreaks::dofix`, `tsDyn::setar`, `tsDyn::lstar`, `MSwM::msmFit`.

## 5.6 Simulation diagnostics

The errors should be i.i.d. Gaussian with mean 0 and should not present any pattern, especially across the parallelization structure. This is guaranteed by the TRNG library, and is checked in Section B.

On top of visualizing the series, to further check for problems in the series generation, the regime-conditional and unconditional moments are estimated and tested against their true values. The regime-conditional true values are calculated as the standard  $AR(1)$  moments. There is only an analytical formula for the unconditional moments of the SB and MS RGP, calculated via iterated expectations.

The table (**TODO?**) in Section B shows the results. Each group of lines corresponds to the moments of a DGP. The first two columns relate to the values conditional on regime 1 and 2, the third column gives the unconditional values. Each cell has the value of the moment, and in brackets the p-value of the null hypothesis that the moment is equal to its true value.

It is expected that models with the same RGP assumption as the DGP return similar moments, so, to check the models' estimation, a similar analysis as above is done in table (**TODO?**). Both diagnoses are generally consistent with the expectations.

To target the focus of this work, the forecast performance (RMSE) was regressed against the index  $i$  of the simulation (Equation 5.2). Table (**TODO?**) shows that the coefficient is not statistically different from zero, even with the high power of the test, as expected.

$$rmse_{p,i,m} = \beta_0 + \beta_1 i + \varepsilon_{p,i,m} \quad (5.2)$$

As a final sanity check, the frequency of improbable events were annotated: TODO% of the datapoints generated were 10 standard deviations away from the relevant mean; and TODO% of the predictions were 10 standard deviations away from the true value.

## 6 Exploratory analysis

Texto de introdução...

### 6.1 Regime separation

There are several aspects in which each DGP's generated series differ, and how the hyperparameters contribute to that difference. I propose that one of the most interesting aspects to analyze is the degree of separation of regimes, in terms of the metrics. There is an important interaction between the sample size  $T$ , the RGP, and the RN that governs that separation, and thus, the models' ability to learn the regime dynamics.

I calculate each metric with the data up to each time-point ( $1 : 2, 1 : 3, \dots, 1 : T$ ). By graphing the last time-point considered on the x-axis, and the difference of the RC metric between regimes on the y-axis, we can see how the separation evolves across sample size.

**Gráfico 1 - separação de regimes e interação com  $t$  e hiperparametrização do RGP e RN:**  $t$  x valor da média das diferenças da métrica entre R1 e R2. Uma cor para cada variação do RGP, uma linetype para cada variação do RN. Um painel (linha) para cada RN, um painel (coluna) para cada métrica. Um Gráfico para cada RGP.

*Análise:* O quão bem separados são os regimes de cada DGP, e como isso dependa da magnitude da diferença definida pela natureza de regime, e da assimetria da ocorrência dos regimes, talvez um pouco também da duração das instâncias (??), e tudo isso interagido com o tamanho amostral  $T$ . E como consideramos isso na hora de criar um conjunto diverso de hiperparametrizações de DGP e  $T$

**Tabela 1 - análise rápida da separação dos regimes em  $T$ :** cada linha um DGP (agrupados por RGP e RN), uma coluna para cada momento, com o valor da medida de diferença/dispersão, sd, e um teste de não zero (ou anova).

*Análise:* Quais os sintomas de cada modelo no que tange a distribuição (momentos) da série.

### 6.2 Regimes and performance

**Gráfico 2 - erros de previsão e acertar regime:** distribuição dos erros de previsão. Cor por acertância. Escolhe variação do RN e RGP. Um painel (linha) para cada RN, um painel (coluna) para cada RGP (talvez tenha que trocar). Um gráfico para cada modelo.

*Análise:* Como acertar o regime importa.

Outras análises possíveis (acho que não vão dar frutos interessantes, vou tentar checar):

- Identificação de parametros e como outros são ajustados para compensar misspecification. Eu acho que esse é muito identificação e não vai trazer grandes intuições sobre essa questão de ajuste.
- Em vez de olhar para o resíduo, dava pra colocar a distribuição do real e do estimado lado a lado (desagregadas por regimes), para ver se existe algum erro sistemático.

## 7 Systematic analysis

Texto de introdução...

**Regressão 1 - dado características, qual modelo vai melhor:** performance  $\sim$  sd(momentos), estratificado por modelo. com modelos da série não predita real e regimes reais. + controles

*Análise: ...*

**Regressão 2 - em qual característica dar match é mais importante:** perf.  $\sim$  diff(sd(carac)), talvez | modelo + controles

*Análise: ...*

**Regressão 3 - identificar cenários que indicam pro econometrista que aumentar ou diminuir S era melhor:** pode ser 1. (e talvez 2.) mas com uma interação de sub- e depois sup- estimação

*Análise: ...*

Outras análises possíveis (acho que não vão dar frutos interessantes, vou tentar checar):

- RGP/RN/modelo (e combinações) e performance. Informação bem pouco útil. Isto é, efeitos fixos de DGP/modelo (tem algum modelo que só é melhor?).
- Interação (RGP, RN) \* modelo e performance. Análise sobre quais interações dão especialmente certo/errado, qual modelo é aproximador universal.
- Identificação e learning:
  - Acertar o r melhor performance?
  - Acertar características do r (duração média, probabilidades de transição, cutoffs, gamma, ...) melhora performance?
  - Acertar os coeficientes melhora a performance?
  - Acertar os sintomas melhora a performance? (o modelo captura a caracterização da série?)
  - Relação fit e performance

## 8 Conclusion

Here I start by summarizing the motivation and methodology.

Then, I focus on the main results. First, with the more descriptive findings about properties of the models, then, the practical recommendations of metrics an econometrician should look at when choosing a model.

## References

- ADAM, Timo; ÖTTING, Marius; MICHELS, Rouven. [Markov-switching decision trees](#). **AStA Advances in Statistical Analysis**, v. 108, n. 2, p. 461–476, 2024.
- AKBAL, O. [Regime-Switching Factor Models and Nowcasting with Big Data](#). **IMF Working Papers**, 2024.
- AKIOYAMEN, Peter; TANG, Yi Zhou; HUSSIEN, Hussien. A hybrid learning approach to detecting regime switches in financial markets. *In: : ICAIF '20.ACM*, out. 2020. Disponível em: <<https://doi.org/10.1145/3383455.3422521>>
- AYDIN, Dursun; MERMI, Selman. [Some Regime-Switching Models for Economic Time Series: A Comparative Study](#). **Eskişehir Technical University Journal of Science and Technology A - Applied Sciences and Engineering**, v. 23, n. 1, p. 48–69, 2022.
- BAI, Jushan; PERRON, Pierre. [Estimating and Testing Linear Models with Multiple Structural Changes](#). **Econometrica**, v. 66, n. 1, p. 47–78, 1998.
- BIERBRAUER, Michael; TRÜCK, Stefan; WERON, Rafał. Modeling Electricity Prices with Regime Switching Models. *In: BUBAK, Marian et al. (orgs.). Berlin, Heidelberg: Springer Berlin Heidelberg*, 2004. Disponível em: <[https://doi.org/10.1007/978-3-540-25944-2\\_111](https://doi.org/10.1007/978-3-540-25944-2_111)>
- BROCKWELL, Peter J.; LIU, Jian; TWEEDIE, Richard L. [On the existence of stationary threshold autoregressive moving-average processes](#). **Journal of Time Series Analysis**, v. 13, n. 2, p. 95–107, 1992.
- CASINI, Alessandro; PERRON, Pierre. **Structural Breaks in Time Series.**, 2018. Disponível em: <<https://arxiv.org/abs/1805.03807>>
- CHANG, Yoosoon; CHOI, Yongok; PARK, Joon Y. [A new approach to model regime switching](#). **Journal of Econometrics**, v. 196, n. 1, p. 127–143, 2017.
- CHEN, Cathy W. S.; LIU, Feng-Chi; SO, Mike K. P. [A review of threshold time series models in finance](#). **Statistics and Its Interface**, v. 4, p. 167–181, 2011.
- CHEN, Dipeng; BUNN, Derek. [The Forecasting Performance of a Finite Mixture Regime-Switching Model for Daily Electricity Prices](#). **Journal of Forecasting**, v. 33, n. 5, p. 364–375, 2014.
- CHIB, Siddhartha. [Estimation and comparison of multiple change-point models](#). **Journal of Econometrics**, v. 86, n. 2, p. 221–241, 1998.
- CHOW, Gregory C. [Tests of Equality Between Sets of Coefficients in Two Linear Regressions](#). **Econometrica**, v. 28, n. 3, p. 591–605, 1960.
- CLEMENTS, Michael P.; KROLZIG, Hans-Martin. [A Comparison of the Forecast Performance of Markov-switching and Threshold Autoregressive Models of US GNP](#). **The Econometrics Journal**, v. 1, n. 1, p. 47–75, 1998.
- DACCO, Robert; SATCHELL, Steve. [Why do regime-switching models forecast so badly?](#) **Journal of Forecasting**, v. 18, n. 1, p. 1–16, 1999.

- DIJK, Dick van; TERÄSVIRTA, Timo; FRANCES, Philip Hans. [Smooth Transition Autoregressive Models — A Survey of Recent Developments](#). **Econometric Reviews**, v. 21, n. 1, p. 1–47, 2002.
- ELIAS, R. S.; WAHAB, M. I. M.; FANG, L. [A comparison of regime-switching temperature modeling approaches for applications in weather derivatives](#). **European Journal of Operational Research**, v. 232, n. 3, p. 549–560, 2014.
- ELLIOTT, Robert J.; SIU, Tak Kuen; LAU, John W. [A hidden Markov regime-switching smooth transition model](#). **Studies in Nonlinear Dynamics & Econometrics**, v. 22, n. 4, p. 20160061, 2018.
- FERGUSON, John D. [Variable Duration Models for Speech](#). In: FERGUSON, John D. (Org.). **Symposium on the Application of Hidden Markov Models to Text and Speech**. [S.l.]: IDA-CRD, 1980. p. 143–179.
- HALL, Anthony D.; PAGAN, Adrian R. [Investigating Some Issues Relating to Regime Matching](#). **Econometrics**, 2025.
- HAMILTON, James D. [A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle](#). **Econometrica**, v. 57, n. 2, p. 357–384, 1989.
- HAMILTON, James Douglas. [22 Modeling Time Series with Changes in Regime](#). In: **Time Series Analysis**. Princeton: Princeton University Press, 2020. p. 677–677.
- HAMILTON, James D.; SUSMEL, Raul. [Autoregressive conditional heteroskedasticity and changes in regime](#). **Journal of Econometrics**, v. 64, n. 1, p. 307–333, 1994.
- HU, Jianchang; SZYMCZAK, Silke. [A review on longitudinal data analysis with random forest in precision medicine](#). **arXiv e-prints**, p. arXiv:2208.04112, ago. 2022.
- JANCZURA, Joanna; WERON, Rafal. [An empirical comparison of alternate regime-switching models for electricity spot prices](#). **Energy Economics**, v. 32, n. 5, p. 1059–1073, 2010.
- KALMAN, Rudolph Emil. [A new approach to linear filtering and prediction problems](#). **Transactions of the ASME – Journal of Basic Engineering**, v. 82, n. Series D, p. 35–45, 1960.
- KIM, Chang-Jin. [Dynamic linear models with Markov-switching](#). **Journal of Econometrics**, v. 60, n. 1, p. 1–22, 1994.
- MEDEIROS, Marcelo Cunha. [Modelos com múltiplos regimes para séries temporais limiares, transições suaves, e redes neurais](#). tese de doutorado—[S.l.]: Pontifícia Universidade Católica do Rio de Janeiro, Departamento de Engenharia Elétrica, 2000.
- PANOPOULOU, Ekaterini; PANTELIDIS, Theologos. [Regime-switching models for exchange rates](#). **The European Journal of Finance**, v. 21, n. 12, p. 1023–1069, 2015.
- PAPARRIZOS, John; YANG, Fan; LI, Haojun. **Bridging the Gap: A Decade Review of Time-Series Clustering Methods**, 2024. Disponível em: <<https://arxiv.org/abs/2412.20582>>
- PINSON, P. *et al.* [Regime-switching modelling of the fluctuations of offshore wind generation](#). **Journal of Wind Engineering and Industrial Aerodynamics**, v. 96, n. 12, p. 2327–2347, 2008.



POTTER, Simon. [Nonlinear Time Series Modeling: An Introduction](#). **SSRN Electronic Journal**, v. 13, 2000.

SONG, Yong; WOŹNIAK, Tomasz. **Markov Switching**. **Oxford Research Encyclopedia of Economics and Finance**Oxford University Press, mar. 2021. Disponível em: <<https://doi.org/10.1093/acrefore/9780190625979.013.174>>

TAN, Zhenni; WU, Yuehua. [On Regime Switching Models](#). **Mathematics**, v. 13, n. 7, 2025.

TERASVIRTA, T.; ANDERSON, H. M. [Characterizing nonlinearities in business cycles using smooth transition autoregressive models](#). **Journal of Applied Econometrics**, v. 7, n. S1, p. S119–S136, 1992.

TERÄSVIRTA, Timo. [Specification, Estimation, and Evaluation of Smooth Transition Autoregressive Models](#). **Journal of the American Statistical Association**, v. 89, n. 425, p. 208–218, 1994.

TONG, H.; LIM, K. S. [Threshold Autoregression, Limit Cycles and Cyclical Data](#). **Journal of the Royal Statistical Society. Series B (Methodological)**, v. 42, n. 3, p. 245–292, 1980.

TONG, Howell. [On a threshold model](#). **Pattern recognition and signal processing**, p. 575–586, 1978.

VERNE, Jean-François. [Smooth threshold autoregressive models and Markov process: An application to the Lebanese GDP growth rate](#). **International Econometric Review (IER)**, v. 13, n. 3, p. 71–88, 2021.

WU, Senlin; CHEN, Rong. [Threshold Variable Determination and Threshold Variable Driven Switching Autoregressive Models](#). **Statistica Sinica**, v. 17, n. 1, p. 241–S38, 2007.

AI disclaimer: this work was generated generally without the help of large language models, except sparringly as a research tool and code autocompletions during the implementation phase.

É necessário colocar algo assim? Talvez como nota de rodapé na primeira página?

## A DGPs, models, and metrics

### A.1 RGPs and models

#### A.1.1 Structural break (SB)

**Hypothesis:** Regime changes at specific time points  $\tau \in (1 : T)^{S-1}$ . Formally:

$$\begin{aligned} r_t^s(\cdot ; \tau) &= \mathbb{1}(\tau'_{s-1} < t \leq \tau'_s), \quad \forall s \in \{1, \dots, S\} \\ \tau &\in \mathbb{N}^{S-1}, \quad \tau_s > \tau_{s-1} \quad \forall s, \quad \tau' = (0, \tau, T) \end{aligned} \quad (\text{A.1})$$

**Empirical model:** Given  $\tau$ , the model estimates  $\mu$  and  $\rho_1$  via OLS in each regime.  $\tau$  is chosen by minimizing the sum of squared residuals over a grid search of breakpoints.

Similarly defined by Bai; Perron (1998). Review of other options by Casini; Perron (2018).

#### A.1.2 Self-exciting threshold (SET)

**Hypothesis:** Regime changes when the series, possibly at a lag  $d \in \mathbb{N}^*$ , crosses specific threshold values  $\tau \in \mathbb{R}^{S-1}$ . Transformations of the variable can be considered<sup>7</sup>. Formally:

$$\begin{aligned} r_t^s(\cdot ; (\tau, d, g)) &= \mathbb{1}(\tau'_{s-1} < g(y)_{t-d} \leq \tau'_s), \quad \forall s \in \{1, \dots, S\} \\ \tau &\in \mathbb{R}^{S-1}, \quad \tau_s > \tau_{s-1} \quad \forall s, \quad \tau' = (-\infty, \tau, \infty), \quad d \in \mathbb{N}^* \end{aligned} \quad (\text{A.2})$$

**Empirical model:** Given  $\tau$  and  $d$ , the model estimates  $\mu$  and  $\rho_1$  via OLS in each regime.  $\tau$  and  $d$  are chosen by minimizing the sum of squared residuals over a grid search of breakpoints and lags. One can also leave  $d$  fixed.

Similarly defined by Tong; Lim (1980). Review of other options by Chen; Liu; So (2011).

#### A.1.3 Smooth transition (ST)

**Hypothesis:** Regime changes smoothly, with a continuous function  $g$ , often a CDF, based on the difference between the series and the threshold  $\tau \in \mathbb{R}$ , possibly at a lag  $d \in \mathbb{N}^*$ . Medeiros (2000) has shown that a generalization to  $S$  regimes is a neural network, but currently, I only consider  $S = 2$ . Formally:

$$\begin{aligned} r_t^1(\cdot ; (\tau, d, g)) &= g(y_{t-d} - \tau), \quad r_t^2(\cdot ; (\tau, d, g)) = 1 - r_t^1(\cdot ; (\tau, d, g)) \\ \tau &\in \mathbb{R}, \quad d \in \mathbb{N}^* \end{aligned} \quad (\text{A.3})$$

Often, the function  $g$  depends on a smoothness parameter  $\gamma$ , i.e., when  $\gamma \rightarrow \infty$ ,  $g \rightarrow \mathbb{1}$ . This parameter can be jointly estimated with the others.

**Empirical model:** Estimated via non-linear squares of the residuals, over  $\mu$ ,  $\rho_1$  (for each regime),  $\tau$ , and  $\gamma$ . Uses some numerical optimization, which depends on starting values and does not guarantee a global optimum.

Similarly defined by Teräsvirta (1994). Review of other options by Dijk; Teräsvirta; Franses (2002)

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<sup>7</sup>For example,  $g(x) = |x|$  or  $g(x) = \Delta x$ .

#### A.1.4 Markov-Switching (MS)

**Hypothesis:** Regime changes stochastically, following a Markov process with transition matrix  $\Gamma \in [0, 1]^{S \times S}$ . The probability of being in regime  $s$  at time  $t$  depends only on the regime at time  $t - 1$ , often with  $\Gamma$  implying some persistence. Formally:

$$\begin{aligned} r_t^s(\cdot; \Gamma) &\sim P(r_t^s = 1 | r_{t-1}) =: \Gamma_{s, r_{t-1}} \\ \Gamma &\in [0, 1]^{S \times S}, \quad \sum_{i=1}^S \Gamma_{s, i} = 1 \quad \forall s \end{aligned} \quad (\text{A.4})$$

**Empirical model:** There are multiple algorithms, including maximum likelihood estimation, expectation maximization, and Markov chain Monte Carlo methods. The EM algorithm uses Kalman to find smoothed probabilities of  $r$ , then the conditional probabilities given the current guess of parameters, then the guess of parameters is updated via maximizing the likelihood given the probabilities. These two steps are iterated until convergence.

Similarly defined by Hamilton (1989). Review of other options by Song; Woźniak (2021).

#### A.1.5 Unsupervised clustering (UC)

**Hypothesis:** no hypothesis on the RGP.

**Model:** Unsupervised clustering techniques, such as K-Means, can be used to estimate the regimes based on  $y_t$ , its lags, and rolling moments. Given the regimes,  $\mu$  and  $\rho_1$  are estimated via OLS. This hybrid approach yields non-standard asymptotic properties. Other clustering techniques could be used, but I focus on the general K-Means clustering problem is as below:

$$\begin{aligned} \hat{r}_t^s(\cdot; (\text{norm}, \text{centroid})) &= \mathbb{1}(y_t \in R_s) \\ R &= \arg \min_{R'} \sum_{s=1}^{\hat{S}} \sum_{y_t \in R'_s} \text{norm}(y_t - \text{centroid}(R'_s)) \end{aligned} \quad (\text{A.5})$$

Similarly defined by Akioyamen; Tang; Hussien (2020). More clustering techniques reviewed by Paparrizos; Yang; Li (2024).

#### A.1.6 Random forests (RF)

**Hypothesis:** there is no RS, the non-linearity is captured by the tree and ensemble structure of the RF. Hu; Szymczak (2022) presents a review of the time series RF literature.

**Model:** a RF is estimated based on  $y_t$ , its lags, and rolling moments.

### A.2 Metrics

The estimated conditional mean and standard deviation can be calculated as, respectively:

$$\begin{aligned} \hat{\mu}(y, r|s) &:= \sum_{t=1}^T r_t^s \cdot y_t \\ \hat{\sigma}(y, r|s) &:= \sqrt{\frac{1}{1 - \sum_{t=1}^T (r_t^s)^2} \sum_{t=1}^T r_t^s \cdot (y_t - \hat{\mu}(y, r|s))^2} \end{aligned}$$

Note the bias correction factor in the denominator of the RC SD, which is necessary given the estimated mean.

As noted, in the case of binary  $r_t$ , only the observations of regime  $s$  have non-zero weights, and the formulas are respectively equivalent to:

$$\frac{1}{|R_s|} \sum_{y_t \in R_s} y_t, \quad \sqrt{\frac{1}{|R_s| - 1} \sum_{y_t \in R_s} (y_t - \hat{\mu}(y|s))^2}$$

For a regime-conditional moments of  $(y_t, y_{t-j})$ , we must define the notion of ‘being in the same regime’. Consider  $r_t^s \cdot r_{t-j}^s$ , which has a correct ‘truth table’ for binary regimes, but also has an interpretation for continuous ones: when closer to 1, the higher the weight of both  $y_t$  and  $y_{t-j}$  being in regime  $s$ . But, this ignores that fact that  $y_t$  and  $y_{t-j}$  can be in the same regime, but in different regime instances. To account for that, the correct weighting should consider the whole window of  $y_{t-j}, \dots, y_t$ :

$$\hat{\rho}_j(y, r|s) = \frac{\sum_{t=1+j}^T \left( \prod_{k=1}^j r_k^s \right) \cdot (y_t - \hat{\mu}(y, r|s)) \cdot (y_{t-j} - \hat{\mu}(y, r|s))}{\sum_{t=1}^T \left( \prod_{k=1}^j r_k^s \right) \cdot (y_{t-j} - \hat{\mu}(y, r|s))^2}$$

Note the absence of bias correction. While it could be present, it can generate larger-than-one correlations, and is often omitted.

For binary regimes, this is equivalent to calculating the unweighted autocorrelation of every concurrent window of regime  $s$ .

Recall that a RC metric returns a sequence with entries for each regime, so when describing the e.g. RC mean  $\mu(y, r)$ , I am referring to:

$$\mu(y, r) := \mu(y, r|S) = (\mu(y, r|S))_{s \in 1:S} \quad (\text{A.6})$$

### A.2.1 True moments of the considered DGPs

Given the weakly stationary within regimes assumption, the regime-conditional moments are independent of the RGP, and are the simple  $AR(1)$  moments:

$$\begin{aligned} \mu(y_t|s) &\equiv E[y_t|y_t \in R_s] = \frac{\mu^s}{1-\rho_1^s} \\ \sigma(y_t|s) &\equiv \text{Var}[y_t|y_t \in R_s] = \sqrt{\frac{(\sigma^s)^2}{1-(\rho_1^s)^2}} \\ \rho_j(y_t|s) &\equiv \text{Corr}[y_t, y_{t-1}|y_t \in R_s] = (\rho_1^s)^j, \quad j \in \mathbb{N}^* \end{aligned} \quad (\text{A.7})$$

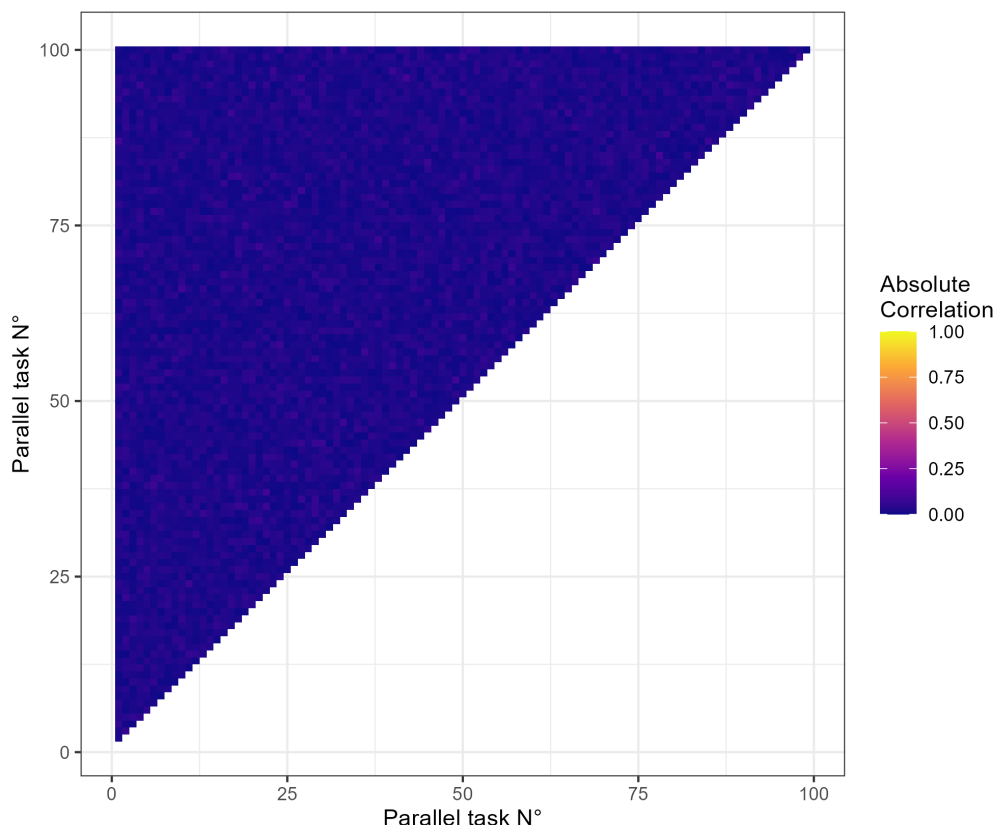
As described in Section 3.3.2, there can be better estimators for populational RC metrics than the ones defined in the above section. One can simply plug in the estimated parameters in equation A.7 to get a better estimator of the moments. In this work, I use this approach, which is also more computationally efficient.

## B Diagnostics

### B.1 Random errors

The Figure 2 shows the correlation of the errors across the parallelization structure. A simple visual check shows no evident patterns and an overall low correlation, as expected.

Figure 2: Errors - Correlation across parallelization structure



The Figure 3 shows the distribution of a size 3000 sample of the errors, via the usual histogram and QQ-plot. The distribution is very close to normal, as expected.

### B.2 Series generation and model estimation

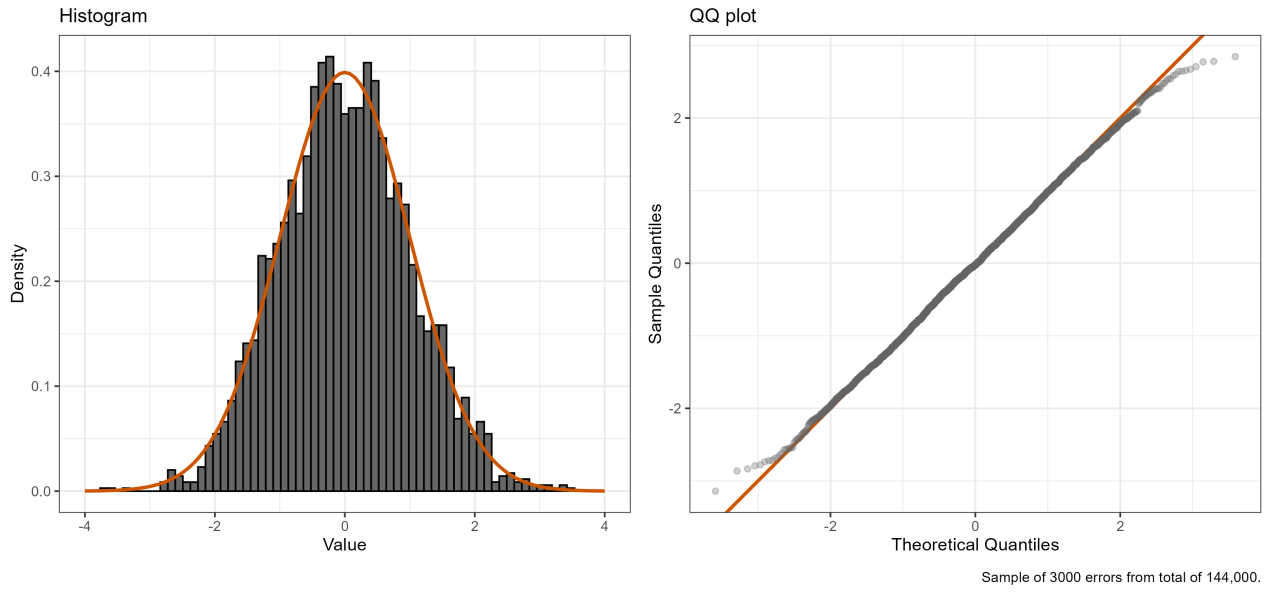
The series diagnostics as described in Section 5.6 are shown in table (TODO?).

**Tabela 1 - séries estão de acordo com momentos analíticos:** cada linha um DGP (agrupados por RGP e RN), um grupo de coluna para cada momento, divididos entre regime 1, 2, e incondicional. Cada célula com o valor do momento e estrelas do teste de igualdade com o momento analítico.

Table (TODO?) is similar as table (TODO?), but shows the estimated moments of the model that matched its line's RGP assumption. The values are the average across simulations of the moments calculated with the estimated parameters. Note that the moments don't need to be exactly the same, since all the models allow for all the parameters to change, a different assumption than that of the regime natures.

Os valores finais dos diagnósticos podem mudar.

Figure 3: Errors - Distribution



| RGP  | RN             | $\mu$   |         |           | $\rho$  |         |           | $\sigma$ |         |           |
|------|----------------|---------|---------|-----------|---------|---------|-----------|----------|---------|-----------|
|      |                | $s = 1$ | $s = 2$ | $\perp s$ | $s = 1$ | $s = 2$ | $\perp s$ | $s = 1$  | $s = 2$ | $\perp s$ |
| RGP1 | $\Delta\mu$    | 9.4*    |         |           |         |         |           |          |         |           |
|      | $\Delta\rho$   |         |         |           |         |         |           |          |         |           |
|      | $\Delta\sigma$ |         |         |           |         |         |           |          |         |           |
| RGP2 | $\Delta\mu$    |         |         |           |         |         |           |          |         |           |
|      | $\Delta\rho$   |         |         |           |         |         |           |          |         |           |
|      | $\Delta\sigma$ |         |         |           |         |         |           |          |         |           |
| RGP3 | $\Delta\mu$    |         |         |           |         |         |           |          |         |           |
|      | $\Delta\rho$   |         |         |           |         |         |           |          |         |           |
|      | $\Delta\sigma$ |         |         |           |         |         |           |          |         |           |

| RGP  | RN             | $\mu$   |         |           | $\rho$  |         |           | $\sigma$ |         |           |
|------|----------------|---------|---------|-----------|---------|---------|-----------|----------|---------|-----------|
|      |                | $s = 1$ | $s = 2$ | $\perp s$ | $s = 1$ | $s = 2$ | $\perp s$ | $s = 1$  | $s = 2$ | $\perp s$ |
| RGP1 | $\Delta\mu$    | 9.4*    |         |           |         |         |           |          |         |           |
|      | $\Delta\rho$   |         |         |           |         |         |           |          |         |           |
|      | $\Delta\sigma$ |         |         |           |         |         |           |          |         |           |
| RGP2 | $\Delta\mu$    |         |         |           |         |         |           |          |         |           |
|      | $\Delta\rho$   |         |         |           |         |         |           |          |         |           |
|      | $\Delta\sigma$ |         |         |           |         |         |           |          |         |           |
| RGP3 | $\Delta\mu$    |         |         |           |         |         |           |          |         |           |
|      | $\Delta\rho$   |         |         |           |         |         |           |          |         |           |
|      | $\Delta\sigma$ |         |         |           |         |         |           |          |         |           |